PROJECT WORK REPORT

Three-Stage Amplifier Design: Nested Miller Compensation (NMC)

Analog IC Design, Frequency Compensation Techniques for Multi-Stage Amplifiers

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Design Tools	Keysight Advanced Design System (ADS), TSMC 0.35 µm CMOS process design kit

1. Project Summary

This report documents the design of a three-stage operational amplifier using Nested Miller Compensation (NMC); a frequency-compensation topology used to stabilize multi-stage amplifiers by using nested feedback capacitors to split the pole frequencies apart, while meeting a target settling time and settling accuracy. The design was carried out at the transistor level in Keysight ADS on a TSMC 0.35 µm CMOS process, and validated through transient, DC, and branch-current simulation.

Work scope covered the full analog design flow: topology selection and comparison against the alternative Reversed Nested Miller Compensation (RNMC) scheme, closed-loop transfer-function derivation, a systematic pole/zero-based sizing procedure, device-level sizing (W/L, gm, ID) for all 11 transistors and 2 compensation capacitors, and simulation verification against the 0.1% (–60 dB) settling-accuracy target.

2. Objective

- Design a 3-stage CMOS amplifier stable under closed-loop feedback while driving a 2 pF load, using a 1.8 V supply.
- Meet a settling-time specification of 25 ns to within 0.1% (–60 dB) settling error.
- Apply Nested Miller Compensation to split the dominant and non-dominant poles and control the right-half-plane (RHP) zero introduced by the Miller feedback path.
- Translate closed-form pole/zero equations into concrete device sizing (transconductances, compensation capacitors, W/L ratios) and verify by simulation.

3. Technical Approach

3.1 Topology

Two established compensation strategies exist for three-stage amplifiers: Nested Miller Compensation (NMC) and Reversed Nested Miller Compensation (RNMC). NMC was selected for this design. In the NMC topology the second stage is non-inverting and the final (third) stage is inverting; two nested Miller compensation capacitors (Cm1, Cm2) exploit the Miller effect to split the closed-loop poles apart, trading bandwidth for phase margin and settling behavior.

The amplifier was implemented as follows:

- Stage 1; NMOS input differential pair (M1–M5) providing the input transconductance gm1.
- Stage 2; Non-inverting common-source stage with an active current-mirror load (M6–M9), providing gm2.
- Stage 3; Inverting output stage (M10, M11), providing gm3 and driving the output/load capacitance CL.
- Compensation; Cm1 and Cm2 form the nested Miller feedback paths that set the closed-loop pole/zero locations.

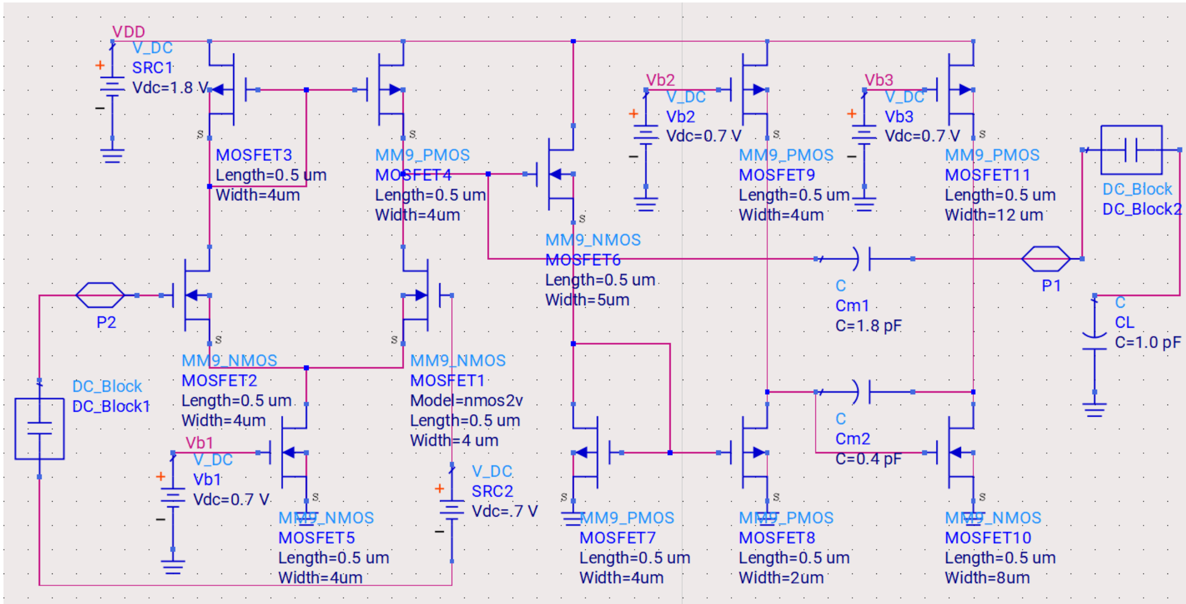

Figure 1;Transistor-level schematic of the three-stage NMC amplifier (ADS / TSMC 0.35 μm)

3.2 Closed-Loop Transfer Function

Under the standard high-DC-gain, CL/Cm1/Cm2 ≫ C1,2, and gm3 ≫ gm1, gm2 approximations, the closed-loop transfer function reduces to a third-order denominator (three poles) with two zeros; one in the right-half plane (RHP) and one in the left-half plane (LHP). The RHP zero sits at a lower frequency than the LHP zero and can erode closed-loop phase margin if not positioned well above the non-dominant pole frequencies; several NMC variants (e.g., adding a series resistor with the compensation capacitors) exist specifically to cancel or push out this RHP zero.

The unity-gain bandwidth was approximated as $\omega GBW \approx gm1/Cm1$, and the pole/zero locations were mapped onto a standardized third-order prototype transfer function (three poles, two zeros) to enable a systematic, closed-form sizing procedure rather than iterative trial-and-error.

3.3 Systematic Sizing Procedure

Design targets were translated into normalized prototype parameters; natural frequency ωn, damping ratio ξ, and normalized zero locations; from which closed-form expressions for gm1, gm2, gm3, Cm1, and Cm2 were derived directly in terms of the load capacitance CL and the settling-time specification. This closed-form mapping (rather than iterative SPICE tuning) is what let the design converge quickly to a device sizing table that already met the settling-time and settling-error targets before layout/simulation refinement.

Design targets used: 0.1% (–60 dB) settling accuracy, settling time tss = 25 ns, load capacitance CL = 2 pF, supply voltage 1.8 V. Solving the settling-error and prototype equations gave $\omega n = 4 \times 10^8$ rad/s with prototype parameters $\alpha = 0.9$, $\xi = 0.85$, and normalized zero locations $z1 = 2$, $z2 = -1.8$.

4. Design Results

4.1 Calculated Circuit Parameters

Solving the closed-form sizing equations yielded the following target small-signal parameters:

Parameter	Target (closed-form)	Used in Design
gm1	≈ 235.3 μA/V	243 μA/V

gm2	≈ 122.8 µA/V	133 µA/V
gm3	≈ 2.33 mA/V	2.26 mA/V
Cm1	≈ 1.535 pF	1.8 pF
Cm2	≈ 0.32 pF	0.4 pF

All active devices were biased in strong inversion with an effective overdrive voltage (Veff) of approximately 200 mV to secure sufficient large-signal output swing and linearity; from the target gm and Veff, bias currents and W/L ratios were derived for every transistor.

4.2 Device Sizing

Device	W/L	gm (µA/V)	ID (µA)
M1, M2	1 × 4 µm / 0.5 µm	243 (gm1)	24.3
M3, M4	4 × 4 µm / 0.5 µm	238	23.8
M5	2 × 4 µm / 0.5 µm	480	48.0
M6	2 × 5 µm / 0.5 µm	133 (gm2)	13.3
M7, M8	1 × 2 µm / 0.5 µm	118	11.8
M9	2 × 4 µm / 0.5 µm	121	12.1
M10	4 × 8 µm / 0.5 µm	2260 (gm3)	226
M11	12 × 12 µm / 0.5 µm	2402	240.2
Cm1	—	1.8 pF	—
Cm2	—	0.4 pF	—

5. Simulation Verification

The completed design was verified in ADS with a 50 mV small-signal step and a 0.8 V large-signal step applied at the input, and with DC operating-point and branch-current analysis across all nodes to confirm correct biasing throughout the signal chain.

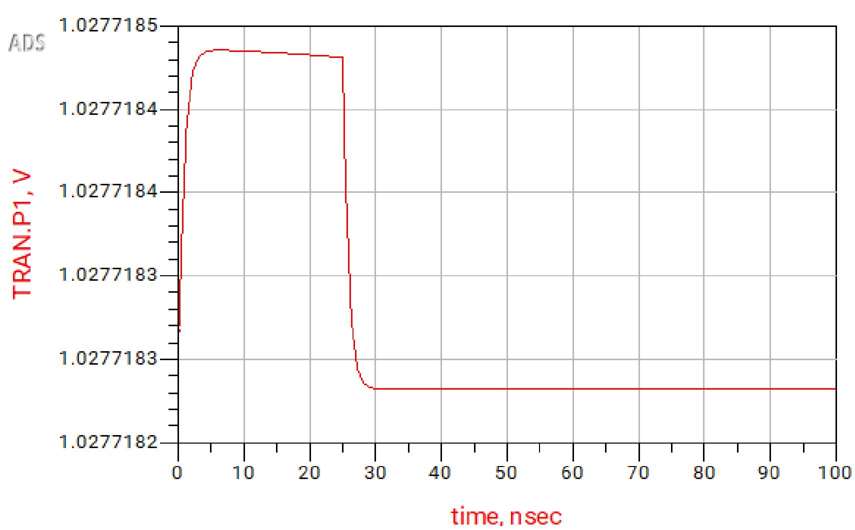

Figure 2;Small-signal transient step response (50 mV input step)

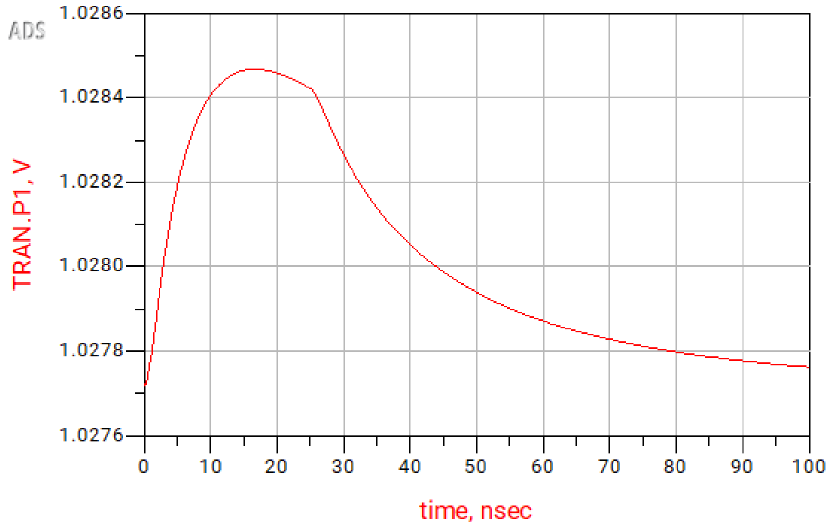

Figure 3;Large-signal transient step response (0.8 V input step)

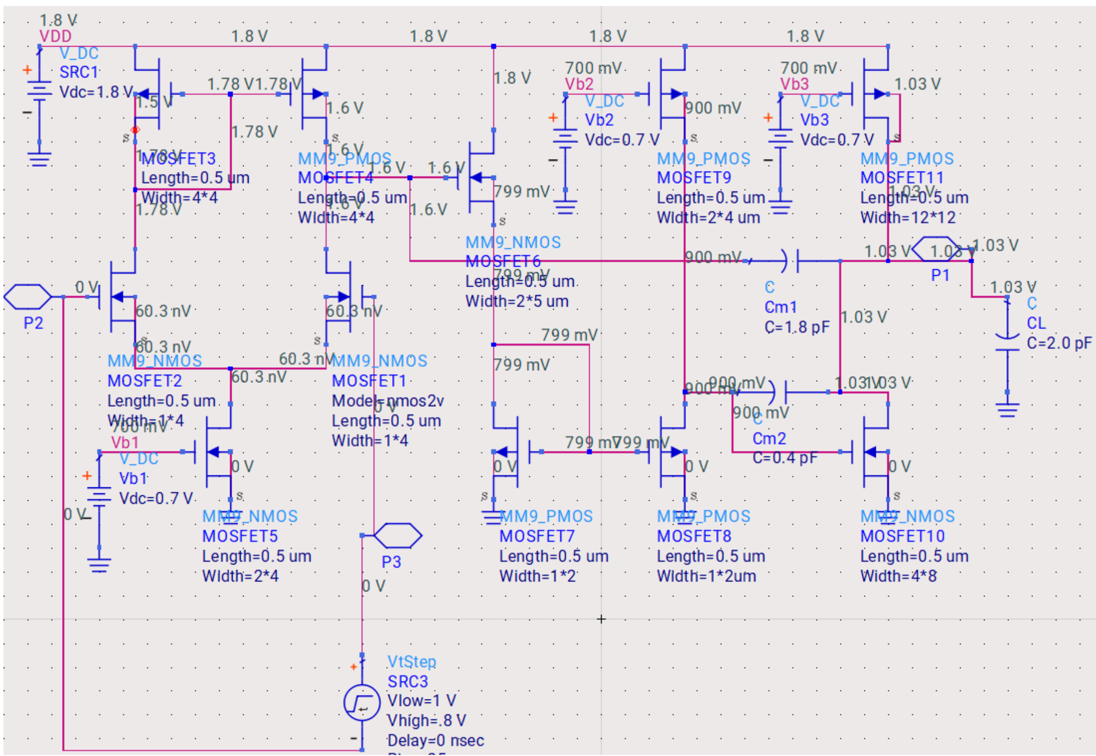

Figure 4;DC operating-point node voltages across the design

The transient responses confirm settled, single-overshoot behavior consistent with the targeted damping ratio (ξ = 0.85) and settling time (25 ns), and the DC/branch-current sweeps confirm all transistors are correctly biased in the intended operating region.

6. Skills Demonstrated

- Analog/mixed-signal IC design: multi-stage CMOS amplifier topology selection and design.
- Frequency compensation theory: pole-splitting, Miller compensation, RHP-zero mitigation in multi-stage feedback amplifiers.

- Analytical circuit design: derivation and application of closed-form pole/zero equations to translate settling-time and accuracy specs directly into device sizing;not just iterative simulation tuning.
- Transistor-level design in an industry process (TSMC 0.35 μm) using Keysight ADS.
- Simulation-based verification: transient (small-signal and large-signal step response), DC operating point, and branch-current analysis.
- Trade-off analysis between bandwidth, phase margin, power consumption, and settling accuracy in feedback-amplifier design.

7. Outcome

The resulting three-stage NMC amplifier met the 0.1% settling-accuracy target within the specified 25 ns settling time on a 2-pF load at a 1.8 V supply, using a fully closed-form, spec-driven sizing methodology validated by transistor-level simulation in ADS.